No-Regret is Not No-Harm: Bandits with Consumable Action Sets

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Abstract

Sublinear regret is the standard justification for deploying sequential decision-making systems. We show that this justification is vacuous when actions consume the action set. We study bandits in which pulling an arm may destroy it, and in which each destroyed arm imposes a permanent negative externality on every subsequent reward. We prove a two-sided separation: a policy achieving *exactly zero* regret against the best-available-arm benchmark can incur value loss $m\delta E$, unbounded in both the pool size m and the externality magnitude δE, and can drive total value negative while the optimal policy remains positive; conversely, any policy near the system optimum is *forced* to report private regret $m\epsilon$. The metric does not merely fail to separate good from destructive policies—it orders them backwards. The mechanism is that regret is measured against an optimum the learner has itself degraded. We further establish that the parameter governing optimal play, the expected marginal externality $\kappa_a = p_a e_a$, is (i) exactly the quantity that determines whether greedy is optimal, and (ii) subject to an irreducible estimation error $\mathrm{SE} \geq 2\sigma/\sqrt{\min(T, \sum_j 1/p_j)}$ that *does not decrease with the horizon*, because each arm supplies exactly one before/after transition and the sample budget is capped by consumption rather than time. The quantity that matters most cannot be estimated arbitrarily accurately, and its attainable precision degrades exactly as it becomes more important. We give a closed-form correction, characterise an exactly-solvable special case, and release an open-source testbed.

1. Introduction

Consider an orchestrator routing tasks across API-backed tools. Some tools are robust; others trip a rate limit, exhaust a quota, or get a key revoked under heavy use, disappearing for the remainder of the session. Tools are not independent: those sharing upstream infrastructure—a vendor, an IP pool, a quota tier—degrade together when one is burned.

Table 1 shows two routers on identical instances. The first has a flawless record: at every step it pulls the highest-value available tool, so its regret against the best-available benchmark is identically zero. It ends on a fallback tool worth 0.470 per call with two tools remaining. The second reports regret of 0.25 per step—and is sitting at 0.782 per call, 66% higher, with four tools remaining.

The greedy router's zero-regret record is not evidence of good behaviour. It is an artefact of a benchmark that degrades in lockstep with the damage being done: every call greedy makes is optimal *given the ecosystem it has already ruined*.

Figure 1 shows the phenomenon in its cleanest form. As the coupling between arms strengthens, the greedy policy's cumulative regret stays pinned at exactly zero while its realised value

Table 1: Two routers, identical seed and tool set, single session. "realised" is the reward actually obtained, $v_{a_t} - B(S_t)$; "regret" is the per-step quantity $\max_{a \in S_t}(v_a - B(S_t)) - (v_{a_t} - B(S_t))$, i.e. instantaneous regret against the best arm available at that step, not a cumulative or expected quantity. The policy that looks worse by the standard metric achieves the better outcome.

router	step	realised	regret	tools left
greedy	0	1.150	0.0000	6
greedy	2	0.844	0.0000	4
greedy	7–13	0.470	**0.0000**	**2**
ECI	0	1.150	0.0000	6
ECI	1–4	0.988	0.0500	5
ECI	5–13	**0.782**	0.2500	**4**

falls monotonically and eventually turns negative—below the value of taking no action at all—even as the optimal policy remains comfortably positive. The two panels are computed on the same instances. A practitioner monitoring the left panel would see nothing wrong at any point.

This paper formalises that failure. Our contributions:

- **Four domain instantiations**—agent tool ecosystems, adaptive cancer therapy under competitive release, platform trials, and CMC design-space development—each simulated, with fit quality stated honestly (Section 9).
- **Theorem 1** (main result). A zero-regret policy can incur value loss $m\delta E$, unbounded in pool size and externality magnitude, and can finish net-negative while the optimum is positive. No-regret is not no-harm.
- **Theorem 2** (converse). Any near-system-optimal policy must incur private regret $m\epsilon$. Preserving value *requires* looking bad on the standard metric, and by Corollary 3 the amount it looks bad by is proportional to the value it delivers.
- **Theorem 4**. Greedy is optimal if and only if $\kappa_a = p_a e_a$ is constant across arms. Neither factor alone is the invariant.
- **Theorem 5**. The externality cannot be estimated to arbitrary precision: the error floor does not decrease with the horizon, because the data-generating process is the consumption itself.
- **Theorem 7**. In the pure-sequencing regime the optimal order is ascending in e; arm values are irrelevant.
- A closed-form correction (ECI), a rollout policy reaching 99.5–99.9% of optimum, and an open-source testbed with one regression test per theorem.

2. Related work

Rotting bandits [4, 5, 6] model rewards that decay with the number of pulls (*rested*) or with elapsed time (*restless*); Seznec et al. [6] give a single adaptive-window policy near-optimal for

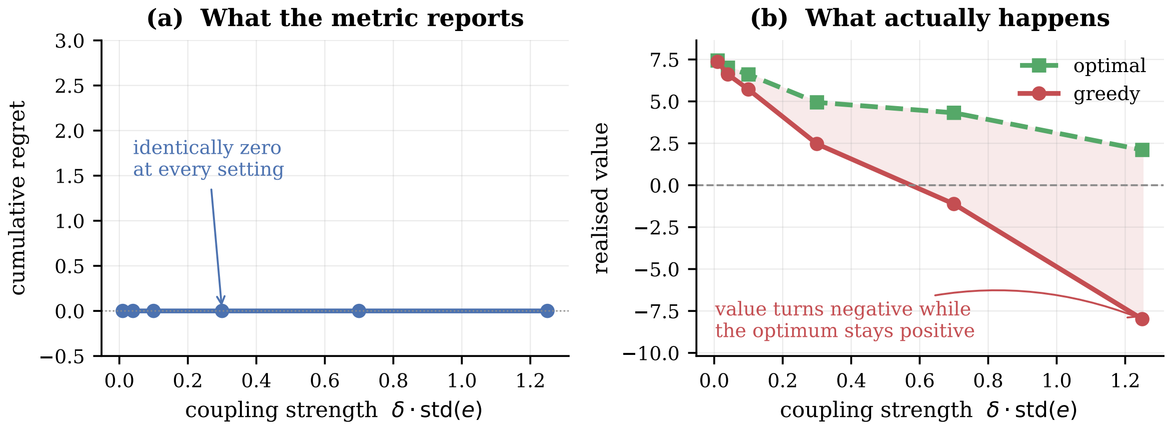

Figure 1: The central phenomenon. **(a)** Greedy's cumulative regret against the best-available-arm benchmark, computed exactly: identically zero at every coupling strength. This is true by construction—greedy maximises the benchmark quantity at each step—and is shown not as a surprise in itself but as the reference against which panel (b) should be read. **(b)** Realised value on the same instances. The shaded region is the loss. At strong coupling the zero-regret policy finishes net-negative while the optimum stays positive. Averages over 20 instances, $n = 6$, $T = 10$, exact dynamic programming.

both. Decay there is gradual rather than removal, and critically the learner is assumed to know which regime it is in. **Mortal bandits** [2] have arms with stochastic lifetimes, but expiry is exogenous. **Blocking bandits** make a pulled arm temporarily unavailable. **Sleeping bandits** [3] allow arbitrary exogenous availability. None of these couple arms: destroying one does not change another's reward distribution.

Bandits with knapsacks [1] is the closest structural neighbour: playing an arm consumes resources and the process halts when a budget is exhausted. The budget is global and exogenous, and consumption leaves other arms' rewards untouched. Here consumption permanently degrades *every* future reward. That difference is why the governing quantity is $\kappa = p \cdot e$ rather than a consumption rate.

Bandit learning with positive externalities [7] is the only prior work we are aware of that places externalities in a bandit model. There, satisfied users attract similar users, so externalities are positive and act on the *arrival process*; the authors show UCB incurs linear regret and develop balanced exploration. Our externalities are negative, act on *rewards*, and are triggered by irreversible destruction. The parallel is worth drawing: in both settings a standard algorithm fails badly, indicating that externality structure of either sign breaks classical guarantees.

Cascading and interacting bandits couple arms within a round—the reward of one depends on which others were displayed or examined—but the coupling is simultaneous and leaves the action set intact across rounds. **Combinatorial bandits** select subsets with structured rewards, again without destroying arms. **Safe and conservative bandits** constrain reward or maintain a baseline while the arm remains available; the constraint restricts *which* actions may be taken, not what taking them removes. **Reinforcement learning with irreversible actions and safe exploration** studies states from which return is impossible, and typically

Table 2: Notation.

symbol	meaning
v_a	mean reward of arm a
p_a	probability that pulling a destroys it
e_a	externality coefficient of a
δ	externality scale
$\kappa_a = p_a e_a$	expected marginal externality of one pull of a
S_t	set of arms available at round t
$B(S) = \delta \sum_{i \notin S} e_i$	accumulated burden
$N_{\text{exh}} = \sum_j G_j,\ G_j \sim \text{Geom}(p_j)$	pulls until the pool is exhausted
$N = \min(T, N_{\text{exh}})$	realised episode length
$W(\pi) = \mathbb{E}_\pi[\sum_t r_t]$	system value of policy π
$V^\star = \max_\pi W(\pi)$	optimal system value
$R_{\text{sys}}(\pi) = V^\star - W(\pi)$	system-value regret
$I(a,t) = v_a - \delta\kappa_a(T-t)$	externality-corrected index (ECI)

seeks to *avoid* irreversibility rather than to price it and schedule it optimally. **Open multi-agent bandits** model a population of learners that arrives and departs exogenously; the churn is in the decision makers, not the actions, and is independent of what is played.

The novelty claim, stated precisely. We do not claim novelty for any one ingredient. Irreversible arm removal appears in mortal bandits; consumption appears in knapsacks; decay with pulls appears in rotting bandits; cross-arm externalities appear in [7]. The combination we study—*(a)* removal that is permanent, *(b)* triggered by the learner's own action, *(c)* carrying a negative externality onto every remaining arm's reward, and *(d)* evaluated against a benchmark that conditions on the already-damaged state—is, to our knowledge, not covered by any of them. It is property (d) combined with (b) that produces Theorem 1: the benchmark and the damage move together. We invite correction on this point and treat it as the claim most in need of scrutiny.

3. Setting

Arms $1, \ldots, n$. Arm a has mean reward v_a, destruction probability $p_a \in (0,1]$, and externality coefficient $e_a \geq 0$. Let S_t be the set of available arms at round t and define the accumulated burden

$$B(S) \;=\; \delta \sum_{i \notin S} e_i.$$

Pulling $a \in S_t$ yields reward $v_a - B(S_t) + \eta_t$ with η_t sub-Gaussian with parameter σ, and destroys a with probability p_a, permanently adding δe_a to the burden. The horizon is T. The value function is

$$V(S,t) = \max_{a \in S} \Big[v_a - B(S) + p_a V(S \setminus \{a\}, t{+}1) + (1-p_a) V(S, t{+}1) \Big].$$

Define the *expected marginal externality*

$$\boxed{\kappa_a \;=\; p_a\, e_a}$$

the expected permanent burden created by one pull of a.

Benchmark. We measure regret against the best available arm, $\max_{a\in S_t}\big(v_a - B(S_t)\big)$. This is the benchmark deployed systems use, and Theorem 1 is a statement about that practice. Value is measured against the exact optimum $V(S_0, 0)$, computed by dynamic programming over subsets for $n \le 8$ and by rollout above that.

4. A zero-regret policy can destroy unbounded value

Theorem 1 (Vacuity of no-regret under consumption)**.** *For any $M > 0$ there is an instance on which greedy attains regret exactly 0 against the best-available benchmark while $V^\star - V_{\text{greedy}} \ge M$. Moreover the loss can exceed $V^\star$, making the zero-regret policy net-negative while the optimum is positive.*

Proof. Fix $m \ge 1$, $E > 0$, $0 < \epsilon < \delta E$. Take $n = m+1$ arms: one *hot* arm H with $v_H = 1$, $e_H = E$, and m *safe* arms with $v_S = 1 - \epsilon$, $e_S = 0$. Set $p_a = 1$ for all arms and $T = n$, so each arm is pulled exactly once and only the order is chosen.

Greedy pulls H first since $1 > 1 - \epsilon$. H is destroyed, adding permanent burden δE, and each of the m subsequent pulls yields $(1 - \epsilon) - \delta E$:

$$V_{\text{greedy}} = 1 + m(1 - \epsilon - \delta E).$$

Deferring H to last yields $1 - \epsilon$ on each safe pull with no burden, and 1 on the final pull of H, whose burden is charged only to later rounds, of which there are none:

$$V^\star \ge m(1 - \epsilon) + 1.$$

Subtracting, $V^\star - V_{\text{greedy}} = m\,\delta E$, unbounded in m and in δE. Choosing $\delta E > 1$ gives $V_{\text{greedy}} < 0 < V^\star$.

At every round greedy pulls $\arg\max_{a\in S}(v_a - B(S))$, which is the definition of the benchmark; its instantaneous regret is 0 at every state. □

The closed form matches exact dynamic programming to machine precision at all 15 settings tested, $m \in \{1, 2, 4, 8, 12\}$ and $\delta E \in \{0.5, 1, 2\}$. At $m = 12$, $\delta E = 2$: $V^\star = 12.88$, $V_{\text{greedy}} = -11.12$, gap exactly 24.0000, regret 0.0000.

Scaling. The gap is linear in the pool size m, not in the horizon T. With $p = 1$ the pool is exhausted after n pulls, so T beyond n is irrelevant; under $p < 1$ the gap grows with T only until $T \approx \sum_j 1/p_j$ and then saturates at the same sample-budget ceiling that appears in Theorem 5. Pool size is the operative quantity because greedy's error is paid once per subsequent pull.

Theorem 2 (Converse: optimality forces private regret)**.** *In the family of Theorem 1, the optimal policy incurs cumulative private regret exactly $m\epsilon$ against the best-available-arm benchmark. Hence for any $M > 0$ there are instances on which every policy within ϵm of the system optimum has private regret at least M.*

Proof. The optimal policy defers H to the final round. At each of the m earlier rounds the best available arm is H (value 1 against $1 - \epsilon$), and the policy declines it, incurring instantaneous private regret ϵ. Summing over the m deferrals gives $m\epsilon$. Any policy that instead pulls H early forfeits δE per remaining round, so for $\epsilon < \delta E$ deferral is necessary for near-optimality. Taking $m \to \infty$ gives the second claim. □

Exact dynamic programming confirms the closed form at all eight settings tested, $m \in \{2, 4, 8, 12\}$ and $\delta E \in \{0.5, 1.5\}$: the optimal policy's private regret is $m\epsilon$ to machine precision while its system-value advantage over greedy is $m\delta E$.

Together with Theorem 1 this closes the separation in both directions. A zero-regret policy may destroy unbounded value, and—the sharper statement—a policy that preserves value *must* report regret. The metric does not merely fail to separate the two; it orders them backwards.

Corollary 3 (The reported regret tracks the value delivered)**.** *In the same family the optimal policy's private regret is $m\epsilon$ and its system-value advantage is $m\delta E$, so their ratio is $\epsilon/(\delta E)$, independent of m.*

Sweeping $\epsilon \in \{0.01, 0.05, 0.10, 0.20, 0.40\}$ at $m = 8$, $\delta E = 1$: the system gain is 8.0000 throughout while private regret runs $0.08, 0.40, 0.80, 1.60, 3.20$—ratios of exactly ϵ.

A second benchmark. Because the choice of benchmark determines the verdict, we report both. Define *system-value regret* $R_{\text{sys}}(\pi) = V^\star - W(\pi)$ against the ex-ante optimal policy. On random instances ($n = 6$, $T = 8$, 15 instances, exact DP):

policy	system value	system regret	private regret
optimal	5.4048	0.0000	0.9489
ECI	5.3315	0.0734	0.6332
greedy	4.4061	0.9987	**0.0000**

The two regret columns rank the three policies in opposite orders. The algorithms are fixed; the benchmark alone determines which appears best.

Interpretation: private versus system value. Define the *system value* of a policy π as its expected total reward, $W(\pi) = \mathbb{E}_\pi\big[\sum_{t=1}^{T} r_t\big]$, so that $V^\star = \max_\pi W(\pi)$, and define the *private value* of an action at state S as its own immediate reward $v_a - B(S)$. Greedy maximises private value at every step. The optimal policy additionally accounts for $\delta\kappa_a(T-t)$—the reward removed from *all subsequent actions* by this one. Regret against the best-available benchmark compares private values only, so it is invariant to that term. Under externalities private and system value diverge, and a guarantee stated in the former says nothing about the latter.

5. When is greedy optimal?

Theorem 4 (Characterisation)**.** *Fix the burden model of Section 3 with additively separable burden.*

- *(i)* (Sufficiency.) *If $\kappa_a = \kappa$ for all arms a, then greedy is optimal at* every *state S and* every *horizon T, for every choice of $\{v_a\}$.*
- *(ii)* (Necessity.) *If $\kappa_a \neq \kappa_b$ for some pair a, b, then there exist $\{v_a\}$ and a horizon T for which greedy is strictly suboptimal.*

The asymmetry in the quantifiers is deliberate and necessary. Constant κ gives *universal* optimality; varying κ gives *existence* of a failure instance, not failure on every instance. Varying

κ with, say, all v_a equal, or a horizon so short that the relevant arms are never reached, need not produce a strict gap. The one-line summary—"greedy is optimal iff κ is constant"—is correct only when read as universal optimality versus existence of a counterexample.

Proof. Write the total value of a policy π over its realised episode as

$$W(\pi) \;=\; \mathbb{E}_\pi\Big[\textstyle\sum_t v_{a_t}\Big] \;-\; \mathbb{E}_\pi\Big[\textstyle\sum_t B(S_t)\Big],$$

and consider the burden term. A destruction at round s imposes δe_{a_s} on every later round of the episode, so

$$\mathbb{E}_\pi\Big[\textstyle\sum_t B(S_t)\Big] \;=\; \delta\,\mathbb{E}_\pi\Big[\textstyle\sum_s \mathbf{1}[\text{destruction at } s]\; e_{a_s}\,(N-s)\Big],$$

where N is the episode length. Conditioning on the history and the arm chosen at round s, the destruction indicator has mean p_{a_s}, so round s contributes $p_{a_s}e_{a_s} = \kappa_{a_s}$ in expectation. Under the hypothesis $\kappa_a \equiv \kappa$ this is the same constant *whichever arm the policy selects*, giving

$$\mathbb{E}_\pi\Big[\textstyle\sum_t B(S_t)\Big] \;=\; \delta\kappa\,\mathbb{E}\Big[\textstyle\sum_s (N-s)\Big] \;=\; \delta\kappa\,\mathbb{E}\Big[\tfrac{N(N-1)}{2}\Big].$$

It remains to note that N is itself policy-independent. Arm j absorbs exactly $G_j \sim \text{Geometric}(p_j)$ pulls before destruction, and every pull is allocated to exactly one arm, so the number of pulls until the pool is exhausted is $N_{\text{exh}} = \sum_j G_j$ *regardless of the order in which a policy allocates them.* Hence $N = \min(T, N_{\text{exh}})$ has a policy-independent distribution.

The burden term is therefore a constant common to all policies, so

$$\arg\max_\pi W(\pi) \;=\; \arg\max_\pi \mathbb{E}_\pi\Big[\textstyle\sum_t v_{a_t}\Big],$$

which is the same problem with no externality at all ($e \equiv 0$). There destruction is geometric, hence memoryless, and rewards are stationary, so an arm's total expected yield v_a/p_a is independent of when harvesting begins: deferring a pull buys no option value and costs the per-round gap. Greedy is optimal in that problem, and therefore in this one, at every state and every horizon.

Necessity. The construction of Theorem 1 has $\kappa_H = E > 0 = \kappa_S$ and gap exactly $m\delta E > 0$, exhibiting the required instance. □ □

The step that closes the argument is that constant κ makes the *entire* expected burden policy-invariant, not merely the per-arm burden term in the Bellman comparison. No coupling argument is needed, and horizon truncation presents no difficulty: the identity holds over the realised episode however long it turns out to be.

Exact verification of the invariance. We confirm the key identity directly by dynamic programming, computing $\mathbb{E}[\sum_t B(S_t)]$ exactly for five structurally different deterministic policies (maximise v, minimise v, maximise p, minimise p, maximise e):

regime	expected burden	spread across policies
constant κ, no exhaustion (T=5, n=10)	0.4000000000	3.3×10^{-16}
constant κ, exhaustion (T=30, n=5)	1.2873273922	2.2×10^{-16}
varying κ (control, T=5, n=10)	0.4444–1.0261	7.3×10^{-1}

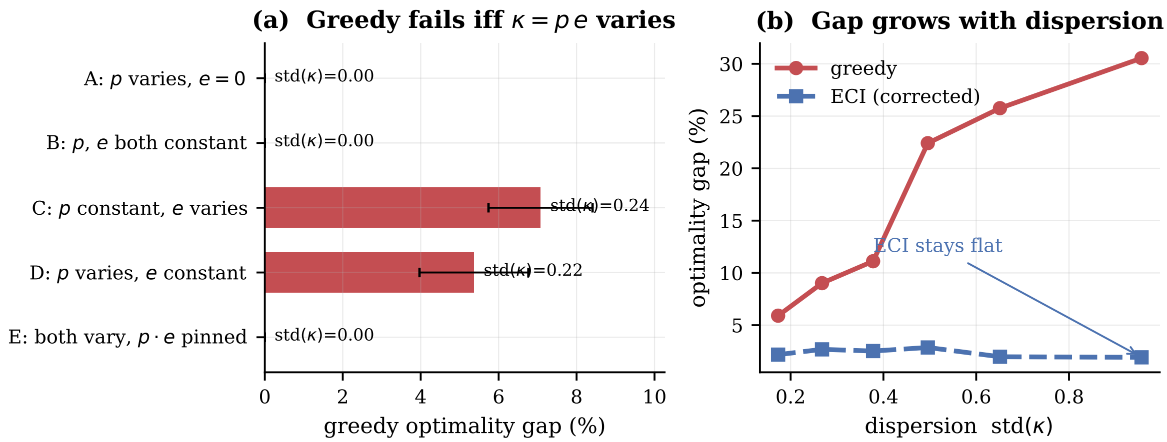

Figure 2: Theorem 4. **(a)** Five structural cases. Greedy is exactly optimal whenever $\kappa = p\,e$ is constant, and fails whenever it is not. Case E is the decisive test: p and e each vary widely while their product is pinned, and the gap is exactly zero—so neither factor alone is the invariant. Cases C and D isolate the two factors separately, and both break greedy. **(b)** The gap grows monotonically with the dispersion of κ, while the corrected index (Section 7.1) stays flat, so its advantage grows without bound. Error bars are standard errors over 12 instances.

Identical to machine precision in both regimes—including the exhausting case, where N is random—and differing by three orders of magnitude when κ varies.

Two empirical checks. Varying p and e widely while pinning κ constant leaves greedy exactly optimal (0/10 instances broken, gap 0.000%); varying the product breaks it (10/10, gap 5.4–6.3%). The theorem also implies the interchange inequality $v_a - v_b \geq p_a D_a - p_b D_b$, which we audit exhaustively as a consistency check: 0 violations in 6,842 ordered pairs under constant κ, versus 56.2% when κ varies.

The optimality gap grows monotonically with std(κ), from 5.9% to 29.1% across six bins over 240 instances. Figure 2 summarises both findings.

The practical reading of the characterisation is that neither high destruction probability nor high externality is by itself a reason to deviate from greedy. An arm that is fragile but harmless, or damaging but robust, can be treated normally. Only the *product*—the damage a pull is expected to cause—distinguishes arms for planning purposes.

6. Irreducible estimation error

Observation model. Rewards are generated by

$$y_t \;=\; v_{a_t} \;-\; \delta \sum_j e_j \,\mathbf{1}[\,j \text{ destroyed before } t\,] \;+\; \eta_t, \qquad \eta_t \sim \mathcal{N}(0, \sigma^2),$$

which is linear in the unknowns $(v_1, \ldots, v_n,\; \delta e_1, \ldots, \delta e_n)$. Because the burden is cumulative, rounds after arm i's destruction generally contain the level shifts of *several* arms, so a pre/post difference in means does *not* by itself isolate δe_i; the parameters must be estimated jointly. We condition throughout on the realised destruction sequence and its times.

Sample budget. Define

$$N_{\text{exh}} \;=\; \textstyle\sum_{j=1}^{n} G_j, \qquad G_j \sim \text{Geometric}(p_j),$$

the number of pulls until every arm has been destroyed, under any policy that eventually exhausts the pool. An episode ends at $N = \min(T, N_{\text{exh}})$, and $\mathbb{E}[N_{\text{exh}}] = \sum_j 1/p_j$.

Theorem 5 (Irreducible estimation error)**.** *Condition on the destruction sequence and let $n_b^{(i)}, n_a^{(i)}$ be the numbers of rounds before and after arm i's destruction, $n_b^{(i)} + n_a^{(i)} = N$. Then for the joint least-squares estimator,*

$$\operatorname{Var}(\delta\hat{e}_i) \;\geq\; \sigma^2\left(\frac{1}{n_b^{(i)}} + \frac{1}{n_a^{(i)}}\right) \;\geq\; \frac{4\sigma^2}{N}, \qquad \textit{hence} \qquad \operatorname{SE}(\delta\hat{e}_i) \;\geq\; \frac{2\sigma}{\sqrt{\min(T,\ N_{\text{exh}})}}.$$

For $T > N_{\text{exh}}$ this bound does not decrease in T.

Proof. Consider the easier problem in which every e_j, $j \neq i$, is *known*. Then the other level shifts can be subtracted from y_t exactly, and the residual process has a single shift δe_i at arm i's destruction time. Its MLE is the difference of pre- and post-destruction residual means, with variance $\sigma^2(1/n_b^{(i)} + 1/n_a^{(i)})$.

In the actual problem the other coefficients are unknown and estimated jointly. Adding free parameters to a linear model weakly increases the variance of every retained coefficient—formally, the design matrix of the oracle problem is a column submatrix of the joint design, and for any $X = [X_1\ X_2]$ the Schur complement gives $[(X^\top X)^{-1}]_{11} \succeq (X_1^\top X_1)^{-1}$. Hence

$$\operatorname{Var}_{\text{joint}}(\delta\hat{e}_i) \;\geq\; \operatorname{Var}_{\text{oracle}}(\delta\hat{e}_i) \;=\; \sigma^2\left(\frac{1}{n_b^{(i)}} + \frac{1}{n_a^{(i)}}\right).$$

By AM–HM, $1/n_b + 1/n_a \;\geq\; 4/(n_b + n_a) \;=\; 4/N$, with equality iff $n_b = n_a$. Finally $N = \min(T, N_{\text{exh}})$, and once T exceeds N_{exh} the binding constraint is the pool rather than the clock. □

The difference-in-means estimator is the MLE only when the other externalities are known; it is invoked here solely as a *lower bound* on the harder joint problem. We verify the inequality chain $\operatorname{Var}_{\text{joint}} \geq \operatorname{Var}_{\text{oracle}} \geq 4\sigma^2/N$ numerically on simulated episodes; it holds in every instance tested. In instances where several arms are destroyed in quick succession the joint design becomes near-collinear and $\operatorname{Var}_{\text{joint}}$ grows by orders of magnitude, which strengthens rather than weakens the conclusion.

All three components verify numerically: the variance formula matches Monte Carlo to within 0.1%; the AM–HM floor is attained exactly at the balanced split (ratio 1.0000); and $\mathbb{E}[N] = \min(T, \sum_j 1/p_j)$ holds including when T binds.

The horizon-independence is stark, and Figure 3 makes it visible. At $p = 0.9$, increasing T from 20 to 320—a sixteenfold increase—changes RMSE by *zero* to four decimal places, because the pool caps the available data at 8.8 rounds no matter how long we wait. At $p = 0.1$ the round count climbs $20 \to 40 \to 71 \to 82.3 \to 83.2$, flattening exactly at the predicted $\sum_j 1/p_j = 80$, and the error curve flattens with it.

The consequence for practice is uncomfortable. In precisely the regimes where irreversible actions matter most—high p, where a single call can permanently remove a resource—the agent has the least opportunity to learn what those actions cost, because every observation is purchased by destroying the thing being priced.

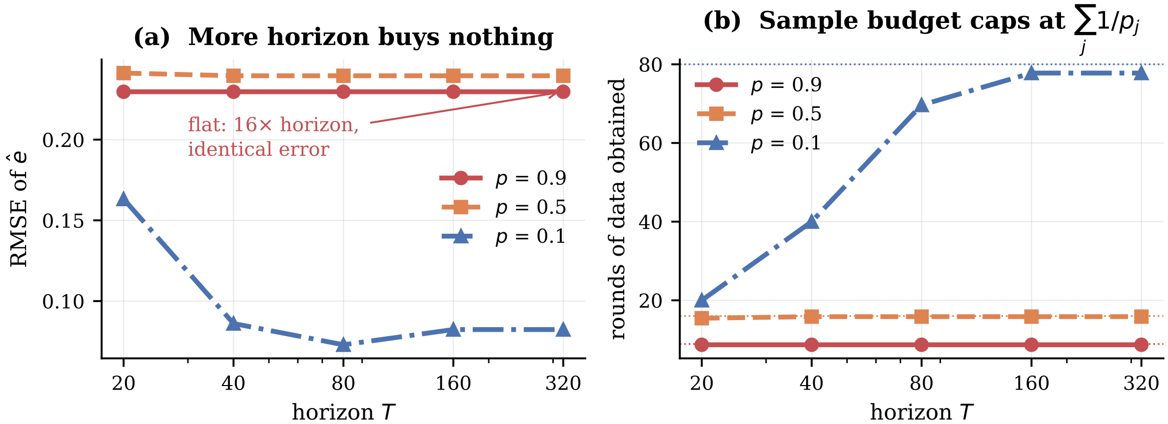

Figure 3: Theorem 5. **(a)** Estimation error for the externality vector as a function of horizon. At high consumption rates the curve is perfectly flat: waiting longer yields no additional information. Only at $p = 0.1$, where the horizon binds before the pool does, does more time help—and only until the pool becomes the binding constraint. **(b)** The mechanism. Dotted lines mark the predicted budget $\sum_j 1/p_j$; the realised number of rounds saturates there. The data-generating process is the consumption itself. Averages over 40 runs.

Corollary 6 (The central tension)**.** *As p rises, $\kappa = pe$ grows, so the externality matters more; simultaneously $N \approx n/p$ shrinks and $\mathrm{SE} \geq 2\sigma\sqrt{p/n}$ degrades. The parameter governing optimal behaviour becomes less estimable exactly as it becomes more important.*

This is not a sample-complexity result that more data resolves. The data-generating process is the consumption itself: observing arm i's externality requires destroying arm i.

Structure reduces but does not remove the difficulty. Suppose $e_a = \langle \psi, z_a \rangle$ for known features $z_a \in \mathbb{R}^k$, $k \ll n$, so each destruction informs a shared k-vector. Per-arm least squares is exactly worthless (identical to greedy to three decimals at every coupling level); feature-based estimation recovers only 17–28% of the available gap; and a *conservative* policy that assumes a uniform upper bound and learns nothing outperforms both at every level. If the mechanism were sample-sharing, capture should rise as k falls. It does not: capture is 4.5% at $k = 1$ and peaks at 30.3% at $k = 5$. At $k = 1$ the externality is nearly uniform, so κ has little dispersion and by Theorem 4 there is nothing to capture. The obstacle is dispersion, not sample size: the externality must vary for the correction to matter, and variation is what makes it hard to estimate.

7. Corrections

7.1 The externality-corrected index

Theorem 4 hands over the correction. Pulling a creates expected permanent burden $\delta\kappa_a$ charged against every remaining round:

$$I(a,t) \;=\; v_a \;-\; \delta\,\kappa_a\,(T-t).$$

Table 3: The regret and value columns are ordered backwards relative to each other.

policy	value	regret
greedy	14.449	**0.000**
Thompson sampling	15.171	1.341
UCB	13.257	6.136
conservative	17.089	8.160
ECI	**21.214**	10.069

Because $B(S)$ is common to all arms it drops out of the ranking, ECI has the decomposition property of an index rule: each arm's score depends only on its own parameters and global time, so the ranking is invariant to accumulated damage. It is an *approximate* index, not an exact one—it omits the option-loss term O_a, it is not derived from a Lagrangian relaxation, and unlike a Gittins index it carries no optimality guarantee. Theorem 7 shows it is not exactly optimal even in the pure-sequencing regime.

Against exact DP, ECI captures 52%–99% of the optimality gap, with capture improving as coupling strengthens. Its own gap is flat at 1.9–3.1% across the entire dispersion range while greedy's grows five-fold, so ECI's advantage grows without bound in std(κ).

7.2 Rollout

No closed form exists for the option-loss term ECI omits. One step of policy improvement over ECI captures it implicitly, since $V_{\pi_0}(S) - V_{\pi_0}(S \setminus \{a\})$ is exactly the marginal value of holding a. This reaches 99.5–99.9% of exact optimum, and with Monte Carlo policy evaluation runs at $n = 40$ where exact DP is infeasible, adding 3.3% over ECI at strong coupling.

7.3 An exactly solvable case

Theorem 7 (Pure sequencing)**.** *If $p_a = 1$ for all a and $T = n$, the optimal order is ascending in e, and arm values are irrelevant to the optimal order.*

Proof. Every arm is pulled exactly once, so $\sum_a v_a$ is a constant. Total value is $\sum_a v_a - \delta \sum_s e_{a_s}(n - s)$: an arm at position s pays its externality $(n - s)$ times. Minimising requires pairing large e with small $(n - s)$. □

Verified by brute force over all $n!$ orderings, exact at 8/8 instances. ECI is *not* exactly optimal even here, and the exact rule inverts the usual intuition: when consumption is certain, an arm's value is irrelevant to scheduling; only its externality matters.

8. Experiments

Table 3 compares policies on 20 arms, 60 rounds, 25 seeds.

Figure 4 extends this across coupling strengths.

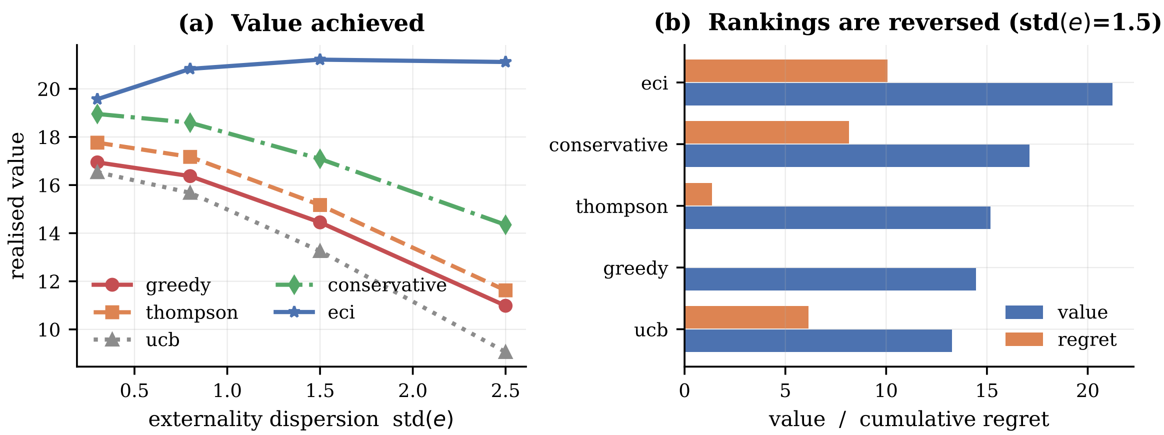

Figure 4: Policy comparison, $n = 20$, $T = 60$, 25 seeds. **(a)** Realised value against externality dispersion. The corrected index dominates throughout, and its margin widens as coupling strengthens. `conservative` uses no per-arm knowledge at all and still beats every consumption-blind method. **(b)** Value and cumulative regret side by side. The two orderings are close to reversed: `greedy` records the lowest regret and near-lowest value; `eci` records the highest regret and the highest value.

Scale and stronger baselines. Comparisons against greedy, UCB and Thompson sampling are weak, since none of them models consumption. We therefore add three methods that do, and run at $n = 50$–100 where exact DP is infeasible: LAGRANGIAN (bandits-with-knapsacks style, maintaining a dual price on expected consumption updated by mirror descent on budget violation), RATIO (maximise $v_a/(1 + p_a e_a)$, the standard resource-aware heuristic), and SAFE-FILTER (exclude arms above a burden quantile, then act greedily).

policy	value (n=100, T=200, std(e)=2.5)	private regret	vs. greedy
ECI	**111.86**	55.73	+52.3%
ratio	111.23	55.12	+51.4%
safe-filter	96.82	57.46	+31.8%
conservative	83.54	34.67	+13.7%
lagrangian	74.86	1.87	+1.9%
greedy	73.46	**0.00**	—
Thompson	71.02	9.30	−3.3%
UCB	69.73	53.32	−5.1%

Two features of the table matter.

The Lagrangian method barely helps (+1.9%), despite being the closest existing approach: it prices consumption, but a scalar dual cannot express the horizon-dependent charge $\delta\kappa_a(T - t)$, which must shrink as the episode runs down while λ cannot. Resource-aware machinery is not sufficient; the correction has to be time-varying.

The ratio heuristic is competitive at this scale, within 1% of ECI. Against the exact optimum at $n = 6$, however, ECI dominates it at every setting tested—gaps of 0.14–2.31% against ratio's 1.27–15.77%—and the margin widens with the horizon, as the $(T - t)$ term predicts. With

Table 4: Domain instantiations. Each row is a running parameterisation, not an illustration.

domain	arm	destruction (p)	externality (e)	fit
Agent tools	API-backed tool	call trips a rate limit, exhausts a quota, or revokes a key	shared upstream infrastructure: burning one throttles the rest	strong
Adaptive therapy	dose level	dose exhausts the drug-sensitive cell population	released resistant clone degrades all later treatment	strong
Platform trial	treatment arm	arm dropped for futility on evidence the allocation generated	reliance on shared control; removal degrades concurrency for all comparisons	moderate
Design space	process parameter setting	run consumes scarce material or yields an out-of-spec batch	failure narrows the defensible filing envelope for all settings	partial

many arms a good low-κ substitute is nearly always available, so the fine-grained weighting matters less; ECI's derivation, rather than its empirical margin at large n, is what the theory contributes.

Thompson sampling is consistently *worse* than greedy under known parameters. Its exploration is not free here: sampling a suboptimal arm can destroy a low-value, high-externality arm. At strong coupling the corrected index nearly triples Thompson's realised value (20.8 vs. 7.2). The mechanisms compose rather than compete—TS handles uncertainty in v, ECI handles consumption—and TS+ECI dominates both.

9. Domains

The model is not specific to agents. We instantiate it in four settings, each supplying its own mapping to (v, p, e, δ) and each simulated rather than merely described. Table 4 gives the mappings, and we state the quality of fit honestly: two are close, one is a reasonable abstraction, and one is partial.

Adaptive therapy is the sharpest case. Under maximum tolerated dose, clinicians administer the highest tolerable dose, which is *exactly* the greedy policy. Higher dose intensity raises immediate tumour control v, but also raises both the chance of exhausting the drug-sensitive population (p) and the damage done when that happens (e), because the sensitive population was suppressing the resistant clone. Since v, p and e rise together, $\kappa = pe$ is highly dispersed—precisely the regime where Theorem 4 says greedy fails. The model therefore reproduces *competitive release* from first principles: it predicts that MTD is beaten by a dose-modulating policy, which is what adaptive therapy trials find. We stress this is a reduced form; real tumour dynamics are a population process, not additive burden from dead arms.

Mechanistic grounding of the other domains. Two further engines derive their parameters from domain dynamics rather than setting them by hand. The *platform-trial* engine models a shared-control trial with staggered entry, in which dropping an arm fragments the control stream into more non-concurrent blocks and every remaining comparison must either discard non-concurrent data or adjust for time drift; v is effect size times information gain, p the futility-crossing probability, and e the precision lost by the remaining arms, weighted by the exiting arm's tenure. The *design-space* engine models process development: v is yield, p the out-of-spec probability rising near the specification limit, and e the fraction of the defensible filing envelope forfeited by a failure there.

A prediction that holds in both directions. On these derived parameters greedy loses 26.8%–121.1% in the tumour engine and 17.3%–106.9% in the design-space engine, with private regret of exactly zero in both. In the platform-trial engine, however, greedy is *optimal*: the gap is 0.0%.

This is not a failure of the theory but a second prediction of it. Dispersion of κ is necessary for greedy to fail, but it must conflict with the value ordering:

engine	std(κ)	corr(v, κ)	greedy
tumour dynamics	0.1705	$+0.614$	fails
platform trial	0.0371	-0.938	optimal
design space	0.2954	$+0.789$	fails

In the tumour and design-space engines, ambition and damage travel together: a higher dose controls more tumour *and* is more likely to exhaust the sensitive compartment; a more aggressive process setting yields more *and* is more likely to truncate the filing envelope. In the platform trial they do not. Promising arms are precisely those that will not cross a futility boundary, so they never fragment the control stream—the externality falls on arms nobody wanted to keep. Value and safety are aligned and greedy is already choosing correctly.

The failure mode is therefore a property of domains in which pushing harder yields more now and costs more later. That is the common case under consumption, but it is not universal, and the platform trial is a clean counterexample the theory predicts correctly.

Results. Figure 5 runs all four. In all four, greedy records cumulative regret of exactly 0.0000 and is beaten—by 30–49% where κ dispersion is largest. Thompson sampling, which is consumption-blind, gains at most 7% and in the agent domain is *worse* than greedy. The conservative policy, which uses no per-arm externality knowledge whatsoever, matches or beats the fully-informed index in three of the four tested domains—the practical form of Theorem 5.

9.1 Agent tool ecosystems in detail

Figure 6 returns to the routing example of Section 1 and shows the three quantities side by side over a single session. Panel (a) is what an operations dashboard displays; panels (b) and (c) are what is actually happening. The greedy router's regret trace is a flat line at zero for the entire session. Its value per call steps downward three times, and its tool count falls to one. The corrected router shows visible, non-zero regret throughout—and ends the session delivering more value per call with more tools still alive.

This is the practical form of Theorem 1: the monitored quantity carries no signal of the loss, so no alerting threshold defined on it can detect the failure.

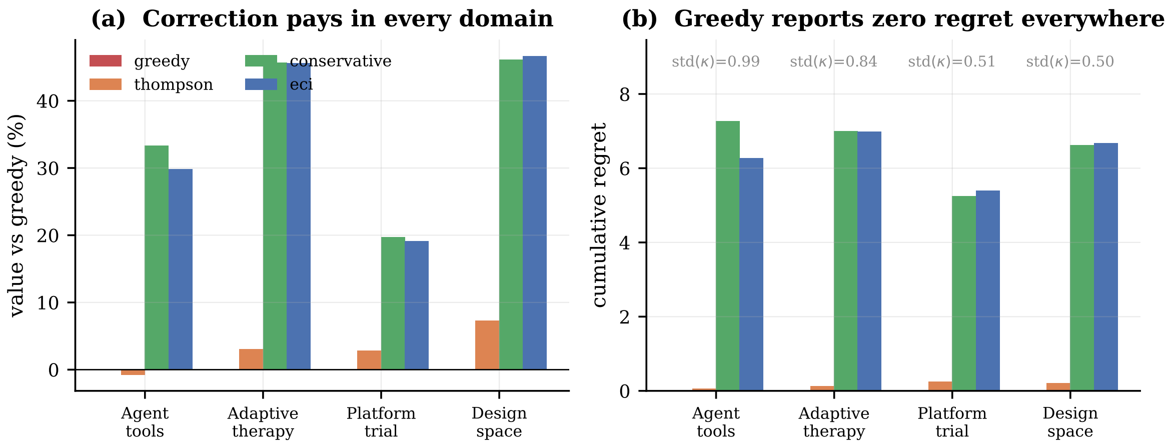

Figure 5: Four domains, 30 seeds each. **(a)** Value relative to greedy. Correction pays in all four tested domains, most where κ dispersion is largest. **(b)** Cumulative regret. Greedy sits at exactly zero in all four domains while losing in all four. The consumption-blind baselines cluster near zero regret with it; the policies that actually perform are the ones the metric penalises.

10. Robustness to the coupling form

The additive burden $B(S) = \delta \sum_{\text{dead}} e_i$ is a modelling choice, so we test it against four alternatives under exact dynamic programming: multiplicative (rewards scaled by $\prod(1-\delta e_i)$), saturating ($B_{\max}(1-e^{-\delta \sum e_i})$), networked (only graph-neighbours of the dead arm are harmed), and concave ($\delta(\sum e_i)^{0.6}$).

Theorem 1 is robust to all five. Greedy records regret of exactly 0.0000000000 under every coupling form while incurring positive value loss under every one (0.49–1.38). This follows from the proof: zero regret is a consequence of greedy's definition against the best-available benchmark, not of the burden's functional form.

Theorem 4 requires separability. Here the picture is more interesting. The characterisation holds exactly for additive, multiplicative and networked coupling (gap 0.000000% under constant κ), and fails slightly for saturating (0.098%) and concave (0.674%). Measuring the marginal burden of destroying one arm at different levels of accumulated damage identifies why: for additive and multiplicative coupling that marginal burden is invariant (0.15000 at zero, three and six units of prior damage), whereas for saturating and concave it falls as damage accumulates ($0.167 \to 0.107 \to 0.068$ and $0.150 \to 0.055 \to 0.043$ respectively).

Corollary 8 (Scope of Theorem 4)**.** *The characterisation holds exactly when the burden is* additively separable *across destroyed arms—when an arm's contribution does not depend on what else has already been destroyed. Under non-separable aggregation, constant κ no longer renders the burden term arm-independent.*

This is a sharper statement than the original, not a weaker one: it names the structural property the result requires. The degradation is also graceful—the residual gap under non-

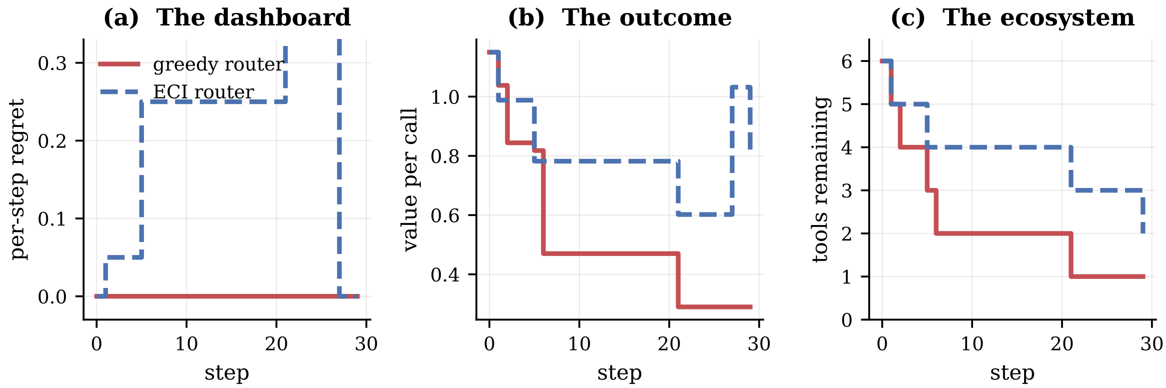

Figure 6: A single routing session, identical seed and tool set. **(a)** The metric: greedy's per-step regret is zero throughout, while the corrected router accrues visible regret. **(b)** Value per call: the router with worse regret delivers more. **(c)** Surviving tools: the router with worse regret preserves more of the ecosystem. The visible regret in (a) is the *price of preservation*, and the standard metric records it as a defect.

separable coupling is 0.10%–0.67%, against the 6.6%–12.2% gap greedy incurs when κ genuinely varies.

11. Multiple decision makers on a shared pool

Consumption is rarely the act of a single learner. Two orchestrators draw on the same API quota; two sponsors compete for the same trial-eligible patients. Each collects its own reward, but a destruction by any of them raises the burden for all. It is natural to expect a commons problem and a corresponding price of anarchy. There is none, and the reason is instructive.

Theorem 9 (Sequential equivalence)**.** *On a shared consumable pool, if agents act one after another within a round and each observes the state left by its predecessors, then m agents over T rounds is exactly one learner over mT pulls.*

Proof. The state is the surviving set together with its accumulated burden. Agent i acting at position j of round t faces exactly the state a single learner faces at pull index $tm + j$; the immediate reward $v_a - B(S)$ is the same function of that state, and the transition—destruction with probability p_a—is identical. The two processes are the same Markov decision process, so any policy in one induces a policy of equal value in the other. □

Verified at five (m, T) combinations: sequential agents and a single greedy learner agree to within 0.015.

Theorem 9 says decentralised greedy agents behave *identically* to one greedy learner, so any price of anarchy computed in this model measures nothing about decentralisation.

Simultaneity does not change the conclusion. Genuine decentralisation requires that agents cannot condition on each other's current choices, so we also implement simultaneous action: all agents choose at once, pulls resolve together, and an arm chosen by several agents faces several independent destruction trials.

m	std(κ)	PoA greedy	PoA ECI	single-agent ratio at mT
2	0.000	**0.9999**	0.9999	1.0000
2	0.635	1.1016	1.0342	1.1916
3	0.000	**1.0029**	1.0052	1.0000
3	0.635	1.1526	1.0567	1.2772

At zero dispersion the price of anarchy is 1.00 even under simultaneity, and with dispersion it sits *below* the single-agent gap at the same total pull count.

Why there is no commons problem. In a classical commons my consumption harms you and not me. Here the burden $B(S)$ is subtracted from *every* agent's reward including my own, on every subsequent pull. The externality is symmetric, and a greedy agent that ignores it entirely behaves identically whether $m = 1$ or $m = 10$. There is nothing to free-ride on when no one is paying.

Corollary 10. *Consumption externalities on a shared pool do not create a price of anarchy. The cost attributable to decentralisation is the single-agent pricing failure, counted once per agent.*

What is genuinely multi-agent. Free-riding is real, because it changes the objective rather than the action space. An agent that prices the externality but discounts it by its own expected share—charging $\delta\kappa_a(T-t)/m$ instead of the full amount—destroys system value: 12.0% at two agents and 13.8% at three, the gap widening as the individual share shrinks. This identifies the target for mechanism design precisely: not the failure to coordinate, which costs nothing here, but the incentive to under-price a cost that is partly borne by others.

Free-riding is charge misspecification. Theorem 9 makes free-riding analysable as a single-learner question. An agent discounting the externality by its own expected share charges $\lambda\,\delta\kappa_a(T-t)$ with $\lambda = 1/m$, so free-riding is ECI with a mis-scaled charge, and the same sweep answers how robust ECI is to any misspecified scale.

Loss is monotone in the discount, and degrades gracefully: even an eightfold discount ($\lambda = 1/8$) recovers 50–63% of the available gain over greedy across settings. Crude partial pricing is worth much more than none.

Two further findings, one negative and one useful.

There is no universal optimal scale. At the base setting $\lambda = 1.5$ beats correct pricing, because ECI omits the option term and so under-prices the true marginal cost. But the minimiser moves from 0.60 at a short horizon to 3.00 at weak coupling, so any specific recommendation would be overfitting.

Over-charging is safer than under-charging, robustly. Comparing a threefold under-charge against a threefold over-charge, over-charging wins in eight of nine settings, by ratios from 3.0 to 20.3; the single exception is a horizon so short that there is little future left for the burden to damage.

Corollary 11. *When the externality scale is uncertain—which by Theorem 5 it always is—err high. Under-pricing costs* $3\times$ *to* $20\times$ *more than over-pricing by the same factor.*

This completes what Theorem 5 leaves open. That result says the scale cannot be estimated; this one says the estimate does not need to be good, provided the error is on the high side. It also explains why the conservative policy of Section 7.1 outperformed every learned estimator: it was not merely robust, it was erring in the safe direction.

12. Terminal commitment

Everywhere above the action set contracts as a side effect of pulling arms. A distinct and practically important variant adds an explicit *terminal action*: after a finite budget of irreversible experiments the learner must declare an operating envelope, and that declaration bounds every subsequent action.

The motivating case is design-space filing under ICH Q8. A finite quantity of active ingredient supports a finite number of process runs; each either demonstrates a parameter setting viable or produces an out-of-spec batch. At filing a design space is declared, and thereafter operating inside it requires no regulatory approval while operating outside triggers a variation procedure. The declared envelope is a permanent commitment made on incomplete evidence.

Model. Settings are indexed $0, \dots, n-1$ with a nominal point at the centre. Yield rises monotonically with process intensity and so does failure probability—the $\mathrm{corr}(v, \kappa) > 0$ structure of Section 9. Each experiment either demonstrates viability or produces an out-of-spec result that blocks that setting permanently. The claimable envelope is the widest contiguous interval of demonstrated settings containing the nominal point. In the operating phase the process drifts; landing inside the envelope collects the yield at the realised setting, landing outside forces reversion to nominal and incurs a variation penalty. Operating value is computed exactly from a discretised drift distribution rather than sampled.

Results (300 seeds per cell):

budget	policy	operating value	envelope width	failed runs
2	greedy (highest yield first)	36.908	**1.00**	0.80
2	edge-first (maximise width)	36.908	**1.00**	0.67
2	**expand outward**	**40.664**	**2.92**	0.08
4	greedy	41.385	3.44	1.06
4	edge-first	36.908 (-10.8%)	**1.00**	1.14
4	**expand outward**	**44.182** ($+6.8\%$)	**4.72**	0.26
6	greedy	45.058	5.35	1.14
6	edge-first	36.908 (-18.1%)	**1.00**	1.36
6	expand outward	45.069	6.20	0.61

Two failure modes appear that have no analogue in the sequential setting.

Greedy buys yield it cannot claim. It spends the budget on the highest-yield settings wherever they lie. But an envelope is an interval containing the nominal point, so a demonstrated setting far from centre is worthless unless everything between is also demonstrated. At budget 2 greedy finishes with envelope width 1.00: it demonstrated the best settings and could claim none of them.

Ambition narrows the envelope it targets. The edge-first policy tests the outermost settings precisely to maximise claimable width. It pays the highest failure probability per experiment, and each failure blocks that setting permanently. Its envelope width is 1.00 at every budget, and its performance degrades monotonically as the budget grows: -10.8% at budget 4, -18.1% at budget 6. Spending more in order to widen the envelope makes it narrower. This is a failure mode practitioners describe directly—an aggressive characterisation campaign ending with a narrower filed design space than a conservative one would have produced.

The budget dependence runs the other way. The advantage of the correct policy *decreases* with budget (+10.2%, +6.8%, +0.0%), the opposite of the sequential setting where the gap grows with pool size. The reason is structural: terminal commitment is a problem about scarcity of evidence at the moment of an irreversible decision. With enough evidence the commitment is no longer made under uncertainty and the difficulty dissolves. It binds hardest exactly when experiments are expensive, which is the regime that motivates it.

13. Software

The accompanying library is released openly and is the artifact through which every claim in this paper can be checked.

Design. The core is dependency-free apart from NumPy. Gym- and PettingZoo-compatible interfaces are implemented natively rather than by importing either framework, so the package carries no reinforcement-learning dependency while matching both signatures.

Observations withhold the externality. By Theorem 5 the externality coefficients cannot be reliably estimated from experience. Exposing them in the observation vector would therefore let an agent solve a different problem from the one the theory describes, so the default observation contains availability, value estimates and pull counts only. A flag lifts this for oracle experiments.

Scenario packs. Seven named scenarios, each documenting the phenomenon it is intended to exhibit, with the test suite asserting that it does so—a scenario that drifts becomes a detectable regression rather than a silently wrong result.

scenario	std(κ)	corr(v, κ)	greedy loss	greedy regret
high-dispersion	0.6285	+1.000	176.5%	0.00000000
design-space	0.2954	+0.789	99.3%	0.00000000
adaptive-therapy	0.1705	+0.614	36.9%	0.00000000
shared-quota	0.9948	+0.315	3.7%	0.00000000
platform-trial	0.0371	−0.938	**0.0**%	0.00000000
aligned-control	0.6299	−0.971	**0.0**%	0.00000000

This table also makes the alignment condition visible in a way the theory alone does not. The `shared-quota` scenario has the *highest* dispersion in the suite and the *smallest* loss, because its correlation is weakest; the synthetic `aligned-control` scenario has larger dispersion than `adaptive-therapy` and yet no loss at all. Across the suite the loss tracks $\mathrm{corr}(v, \kappa)$ rather than $\mathrm{std}(\kappa)$.

Reproducibility. Every figure regenerates from the library and the exact dynamic-programming reference through a single command. All random seeds are fixed in code. The test suite contains one regression test per theorem, so breaking a theorem's numerical claim breaks a test. Experiments that produced negative results are retained in the repository rather than removed.

14. Limitations

Exact optimality results rest on dynamic programming at $n \leq 8$; larger-scale claims rest on rollout, and the coupling-robustness study of Section 10 inherits the same ceiling. Theorem 1 is a statement about the best-available-arm benchmark specifically—a forward-looking benchmark would not be fooled, but it is also not what deployed systems measure, which is the point. The domain instantiations vary in fidelity. The agent and adaptive-therapy mappings are structurally faithful; the platform-trial mapping treats an inferential externality (loss of concurrent control) as if it were a reward penalty, and mixes patient benefit with information gain in a single scalar; the design-space mapping omits the terminal-commitment layer that defines the real problem—a one-shot declaration of design-space width that fixes the feasible set permanently. That layer is an optimal-stopping problem on top of the bandit and is, in our view, the most promising direction this work opens.

15. Conclusion

When actions consume the action set, regret is measured against an optimum the learner has already degraded. A policy can hold regret at exactly zero while driving value arbitrarily negative, and the parameter that would let it do better admits an irreducible estimation error that worsens precisely as it becomes important, because acquiring each observation destroys the resource being priced. The practical consequence is direct: for irreversible actions, externality costs must be *specified*, not discovered. A crude uniform upper bound outperforms every learned estimator we tested.

Code and reproducibility. The library, all 32 experiments including negative results, and one regression test per theorem are released openly. Every figure regenerates from the library and the exact-DP reference via a single command, all random seeds are fixed and stated in code, and the test suite fails if any theorem's numerical claim breaks. **[Repository URL to be inserted before posting.]**

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